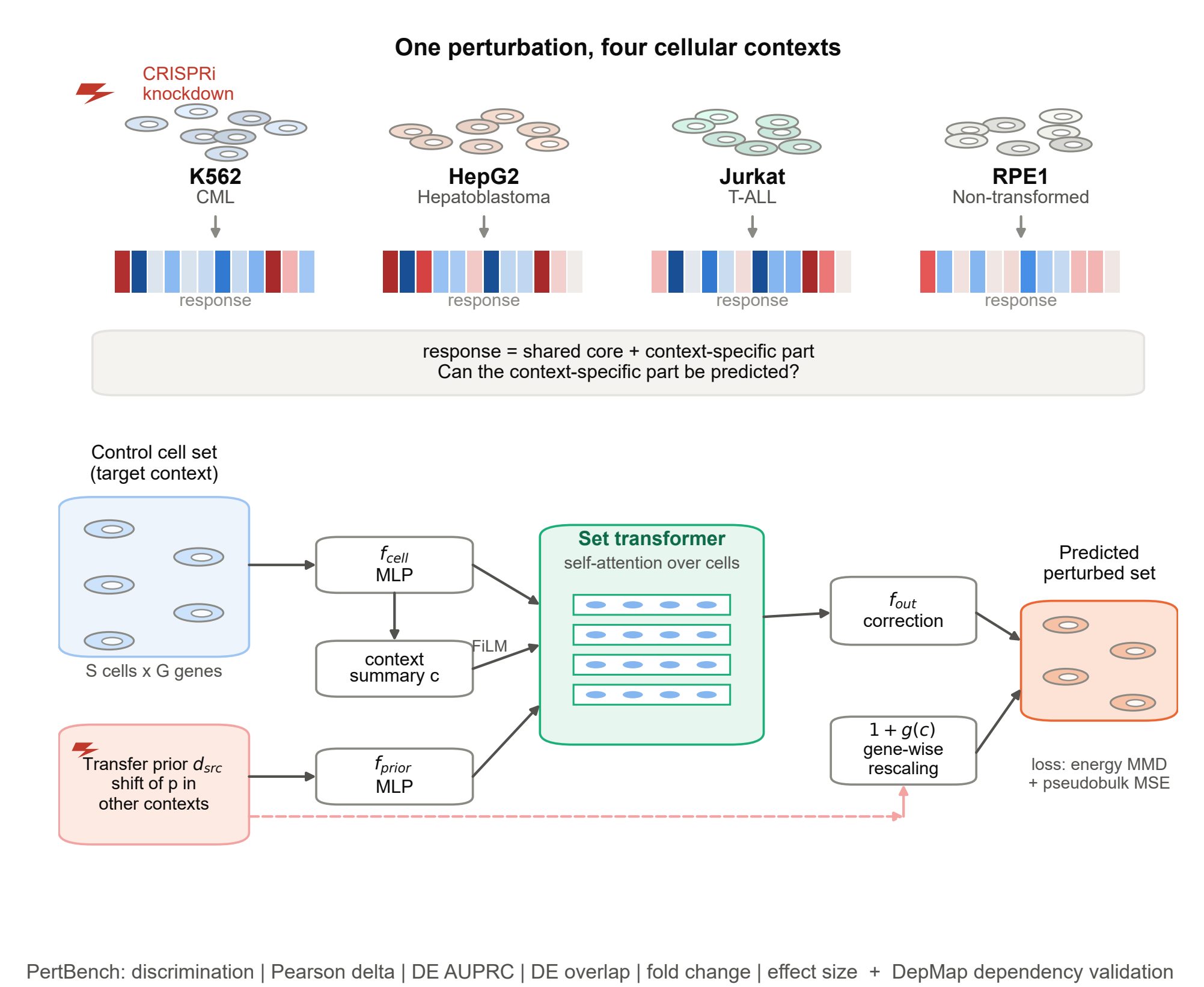

One perturbation, four cellular contexts
CRISPRi knockdown
K562
CML
HepG2
Hepatoblastoma
Jurkat
T-ALL
RPE1
Non-transformed
response
response
response
response
response = shared core + context-specific part
Can the context-specific part be predicted?
Control cell set (target context)
S cells x G genes
f_{cell} MLP
context summary c
FiLM
Set transformer
self-attention over cells
f_{out} correction
Predicted perturbed set
Transfer prior d_{src} shift of p in other contexts
f_{prior} MLP
$1 + g(c)$ gene-wise rescaling
loss: energy MMD + pseudobulk MSE
PertBench: discrimination | Pearson delta | DE AUPRC | DE overlap | fold change | effect size + DepMap dependency validation